Mastering Generative AI & LLM Engineering
From Foundations to Applications

Matplotlib
Figure Object

PART ONE : UNDERSTANDING THE FIGURE

Led by : Shrevas Shukla
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The more comprehensive Matplotlib OOP API makes use of a Figure object.

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We then add axes to this Figure object and then plot on those axes. This allows for very robust controls over the entire plot.

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Let’s quickly visually build an understanding of the Figure object before coding it with Python…

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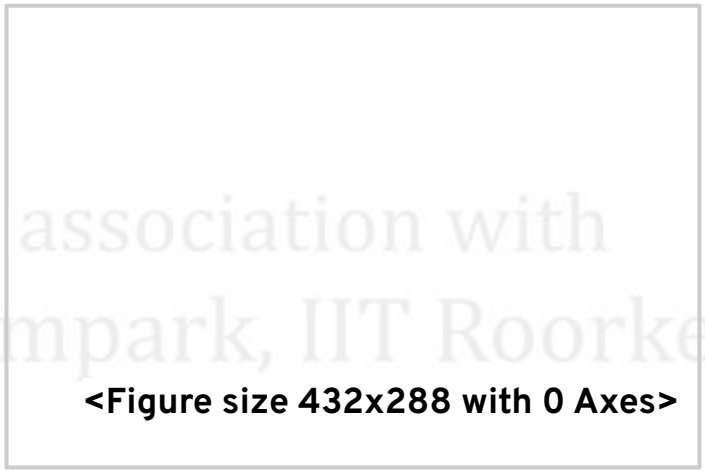
Note: The Figure object we’re about to show is technically not visible until you add axes to it.

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plt.figure()

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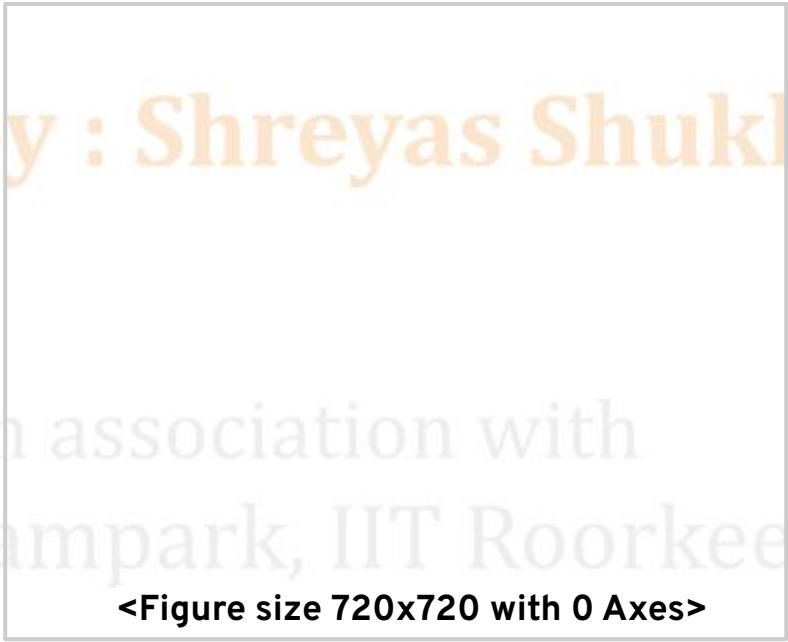


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```
plt.figure(figsize=(10,10))
```

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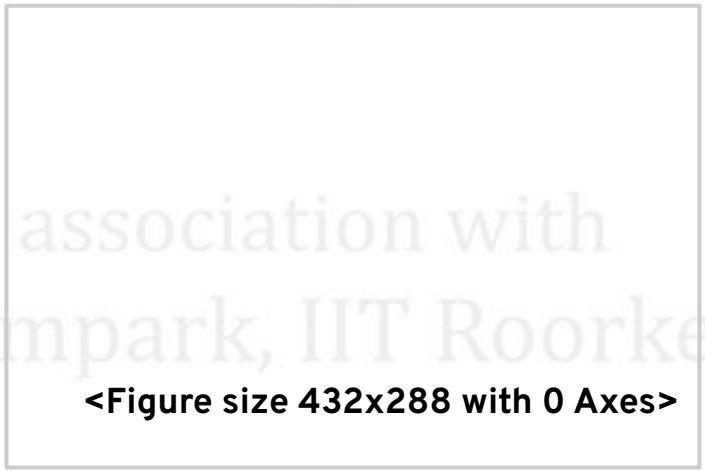


```
<Figure size 720x720 with 0 Axes>
```

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```
fig = plt.figure()
```

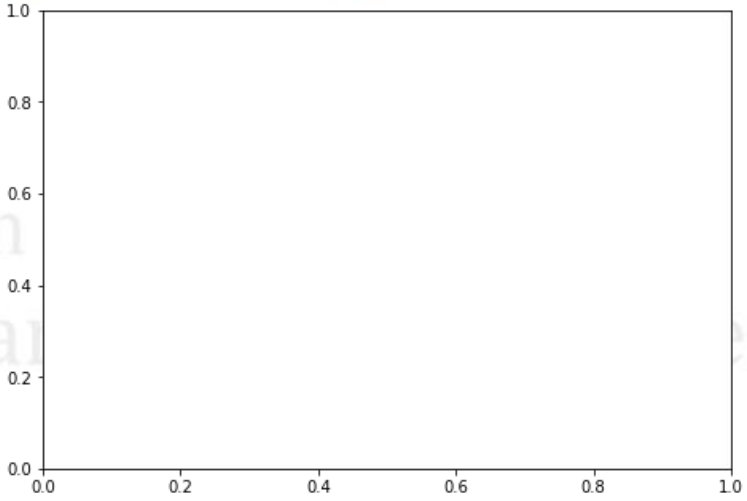
Blank canvas, waiting for a set of axes for plotting.



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```
fig = plt.figure()

axes = fig.add_axes([0, 0, 1, 1])
```



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```
fig = plt.figure()

axes = fig.add_axes([0, 0, 1, 1])
```

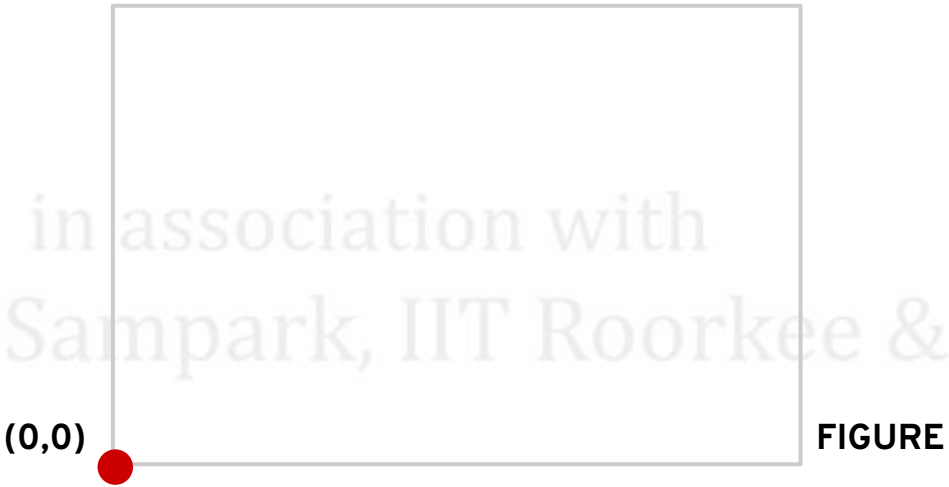
Lecture by: Sanjeyas Shukla
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FIGURE

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- fig = plt.figure()
  axes = fig.add_axes([0, 0, 1, 1])

(x,y)
Lower Left Corner of Axes

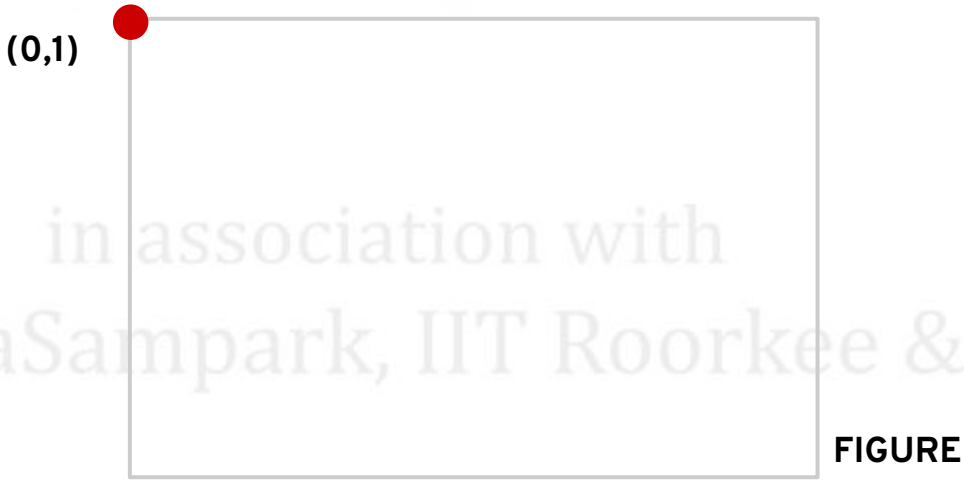


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- fig = plt.figure()
  axes = fig.add_axes([0, 1, 1, 1])

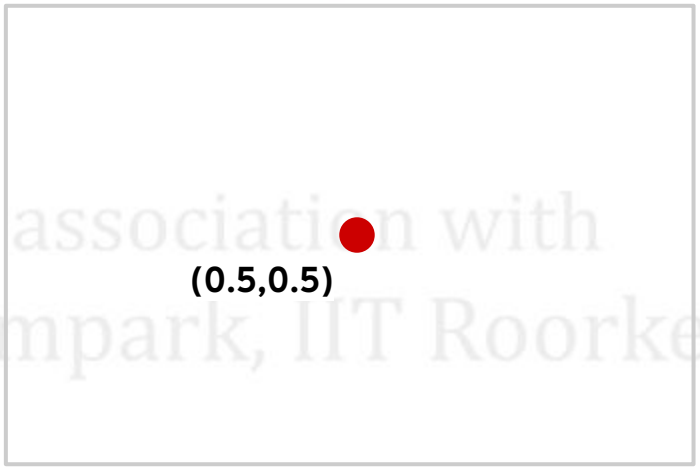
(x,y)
Lower Left Corner of Axes



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- fig = plt.figure()
  axes = fig.add_axes([0.5, 0.5, 1, 1])

(x,y)
Lower Left Corner of Axes



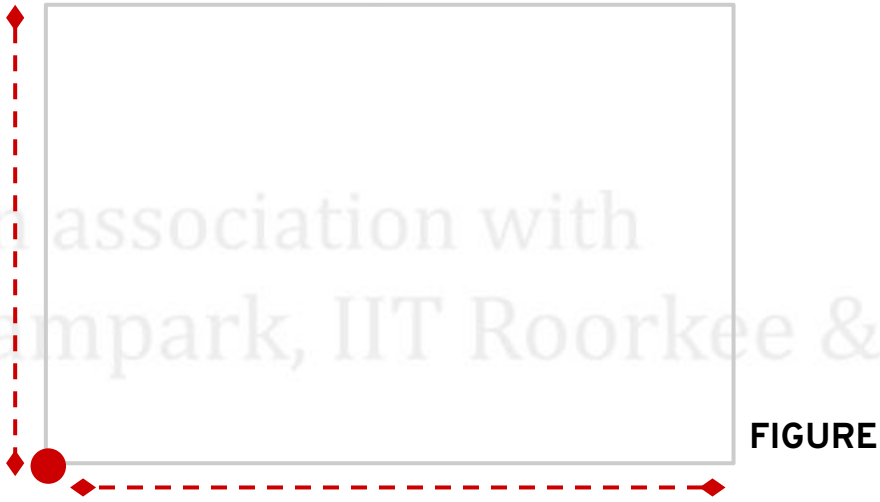
FIGURE

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- fig = plt.figure()
  axes = fig.add_axes([0, 0, 1, 1])

(width,height)
of Axes

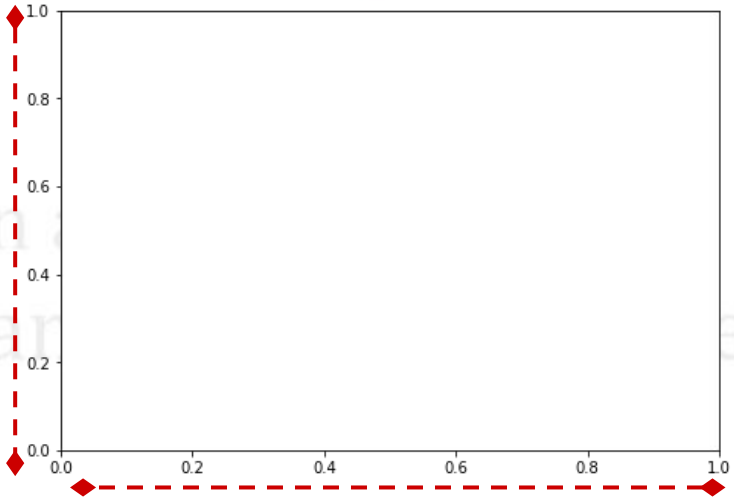


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- fig = plt.figure()
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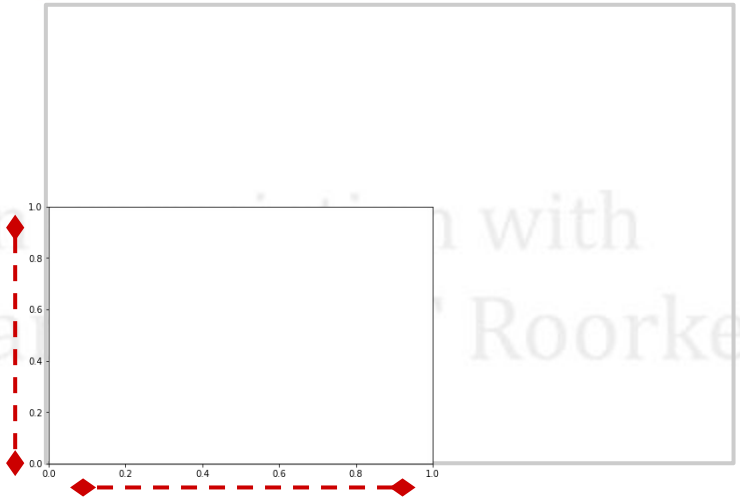
(width,height) of Axes



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- fig = plt.figure()
  axes = fig.add_axes([0, 0, 0.5, 0.5])

(width,height) of Axes



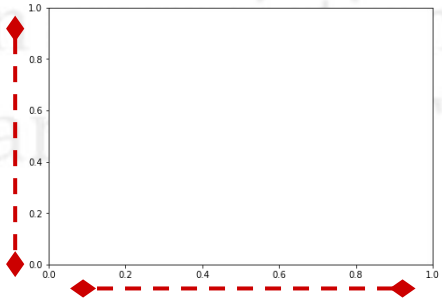
FIGURE

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- fig = plt.figure()
  axes = fig.add_axes([0, 0, 0.5, 0.5])

(width,height)
of Axes

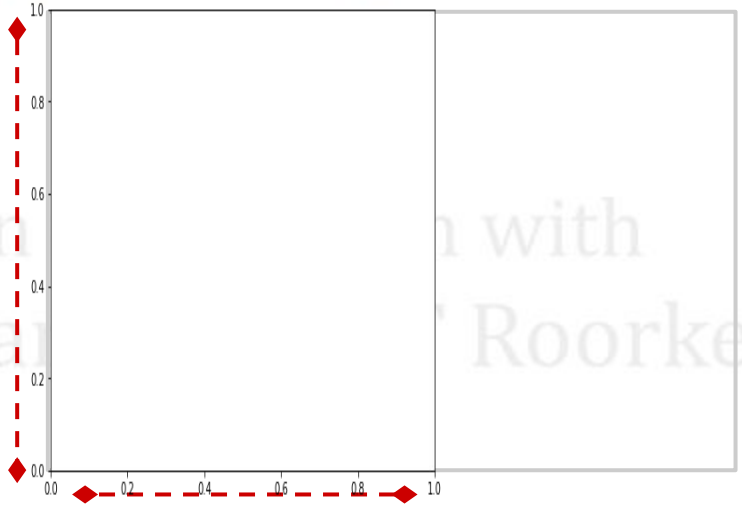
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- fig = plt.figure()
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(width,height)
of Axes

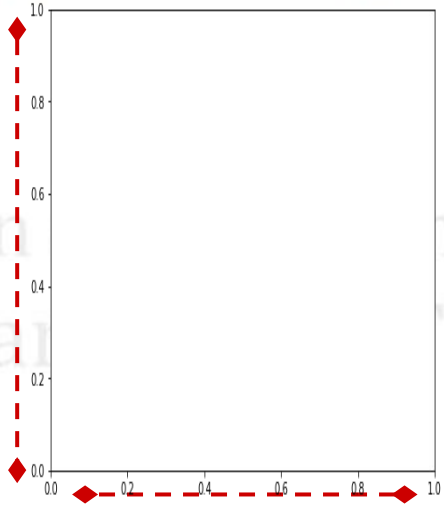


FIGURE

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- fig = plt.figure()
  axes = fig.add_axes([0, 0, 0.5, 1])

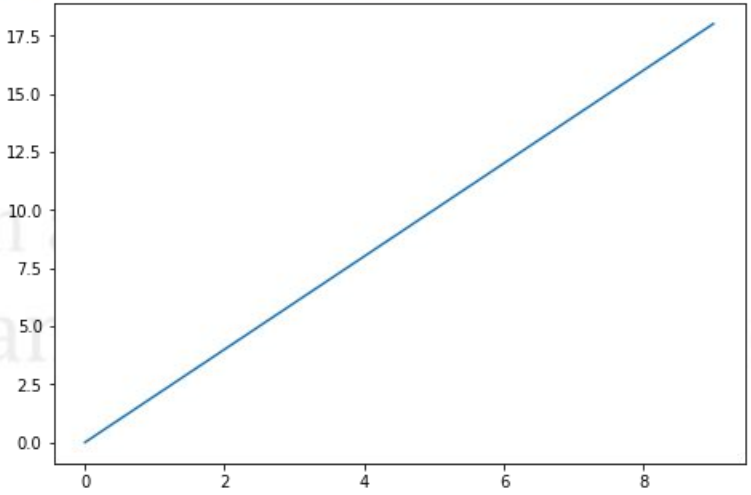
(width,height) of Axes



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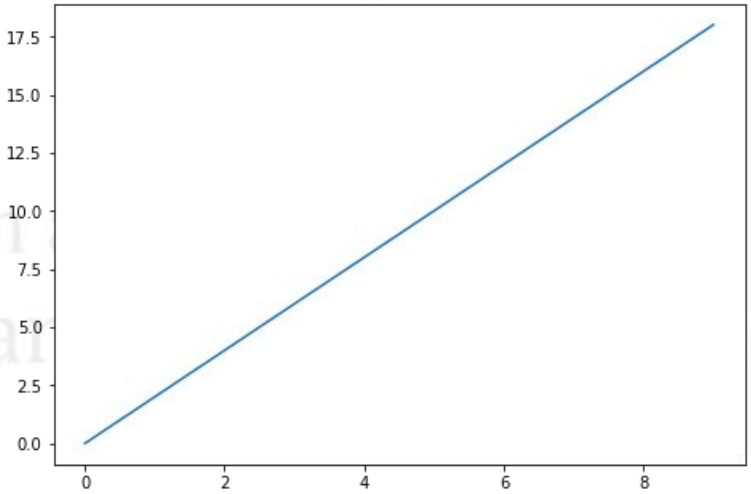
- fig = plt.figure()
  axes = fig.add_axes([0, 0, 1, 1])
  axes.plot(x, y)

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- fig = plt.figure()
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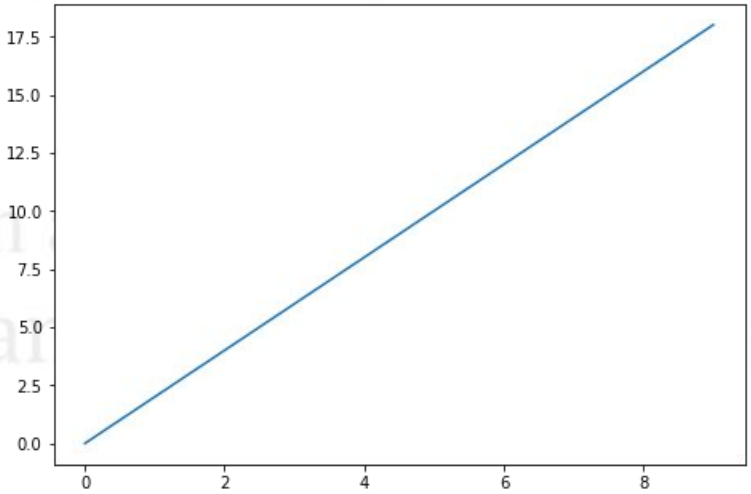


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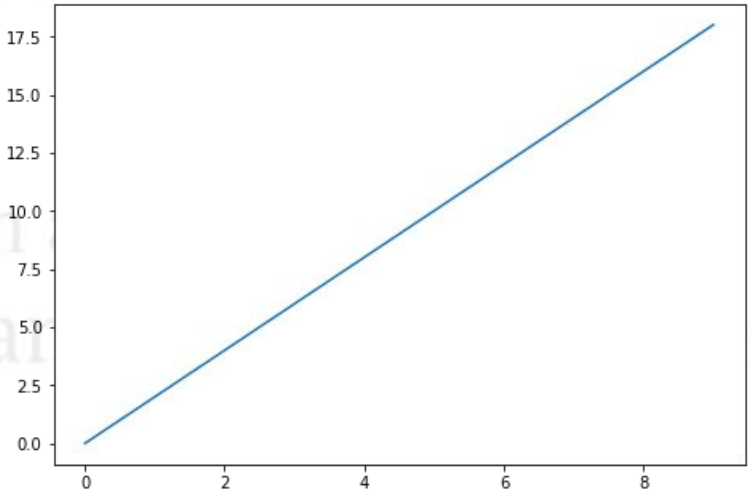
- fig = plt.figure()
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- fig = plt.figure()
  axes = fig.add_axes([0, 0, 1, 1])
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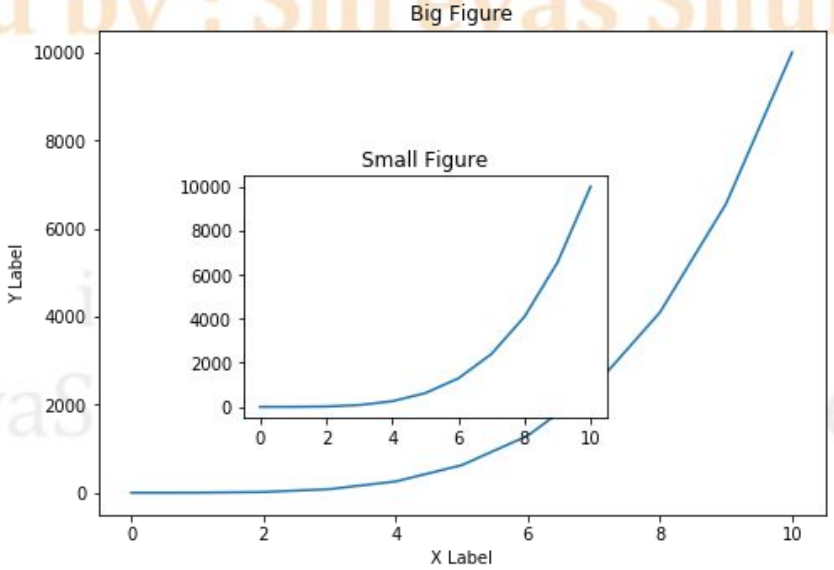


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This methodology allows us to add in multiple axes as well as move and resize the axes.

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In theory we could set axes side by side using plt.figure() calls, but typically it is easier to use plt.subplots() function calls for this.

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Matplotlib SubPlots

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In theory we could create a Figure object and then manually add and arrange sets of axes to line up multiple plots side by side.

However, Matplotlib comes with a pre-configured function call **plt.subplots()** that automatically does this for us!

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The plt.subplots() call allows us to easily create Figure and Axes objects in side by side formations.

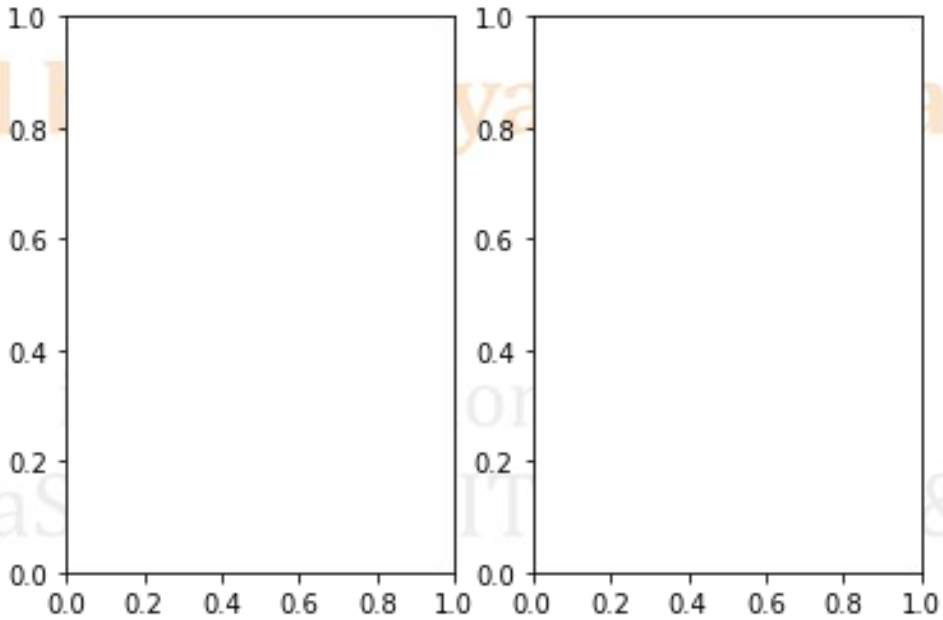
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The plt.subplots() command returns a tuple containing the Figure canvas and then a numpy array holding the axes objects.

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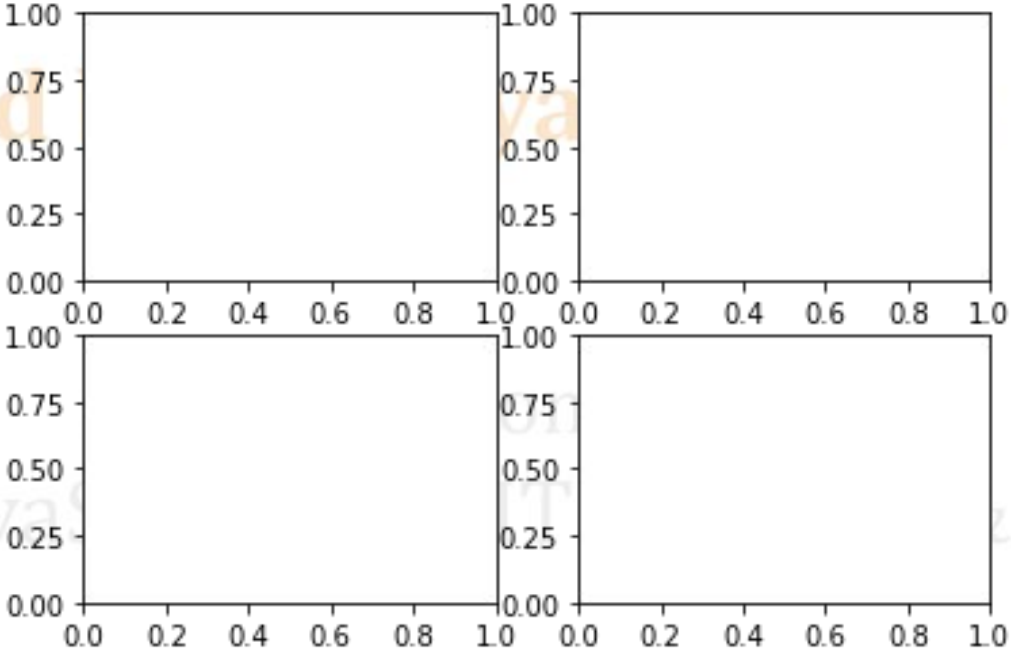
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- fig, axes = plt.subplots(nrows=1, ncols=2)



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- fig, axes = plt.subplots(nrows=2, ncols=2)



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plt.subplots() returns a tuple which by common convention we label **(fig,axes):**

- **fig**
  - This is the entire Figure canvas.
- **axes**
  - This is a numpy array holding each of the axes according to position in the overall canvas.

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Let’s explore how to use plt.subplots() to easily create and align multiple plots!

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Natural Language Processing

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Topics for this Module

1. Exploration and Application
2. Preprocessing
3. Speech Tagging and Named Entity Recognition
4. Sentiment Analysis
5. Text Vectorization
6. Topic Modeling
7. Custom Classifier
8. Real World Application

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1950 : First traces of NLP



1990 : Booming Development of the Web



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2012 : Availability of Huge Datasets

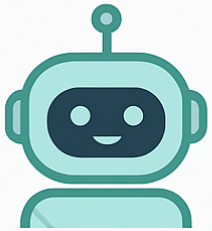


2022 : ChatGPT

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NLP in everyday life

1. Search engines use NLP to interpret user intent and return accurate results.
2. Spam detection systems rely on classification algorithms to filter unwanted emails
3. Chatbots use NLP to understand and respond to user queries.



Shreyas-Assistant

Q: Who are you?

I'm Shreyas-Assistant, an AI education assistant built by Shreyas Shukla to help you learn AI, Machine Learning, Deep Learning, and Python.

A: What can you do?

I can explain AI concepts, guide you through ML workflows, help debug Python code, suggest learning paths, and share insights from Shreyas's courses and YouTube channel

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Supervised and Unsupervised

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Data Preprocessing

- General Cleaning
- Removing Noise
- Getting the Data in the right format

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Lets Code !!